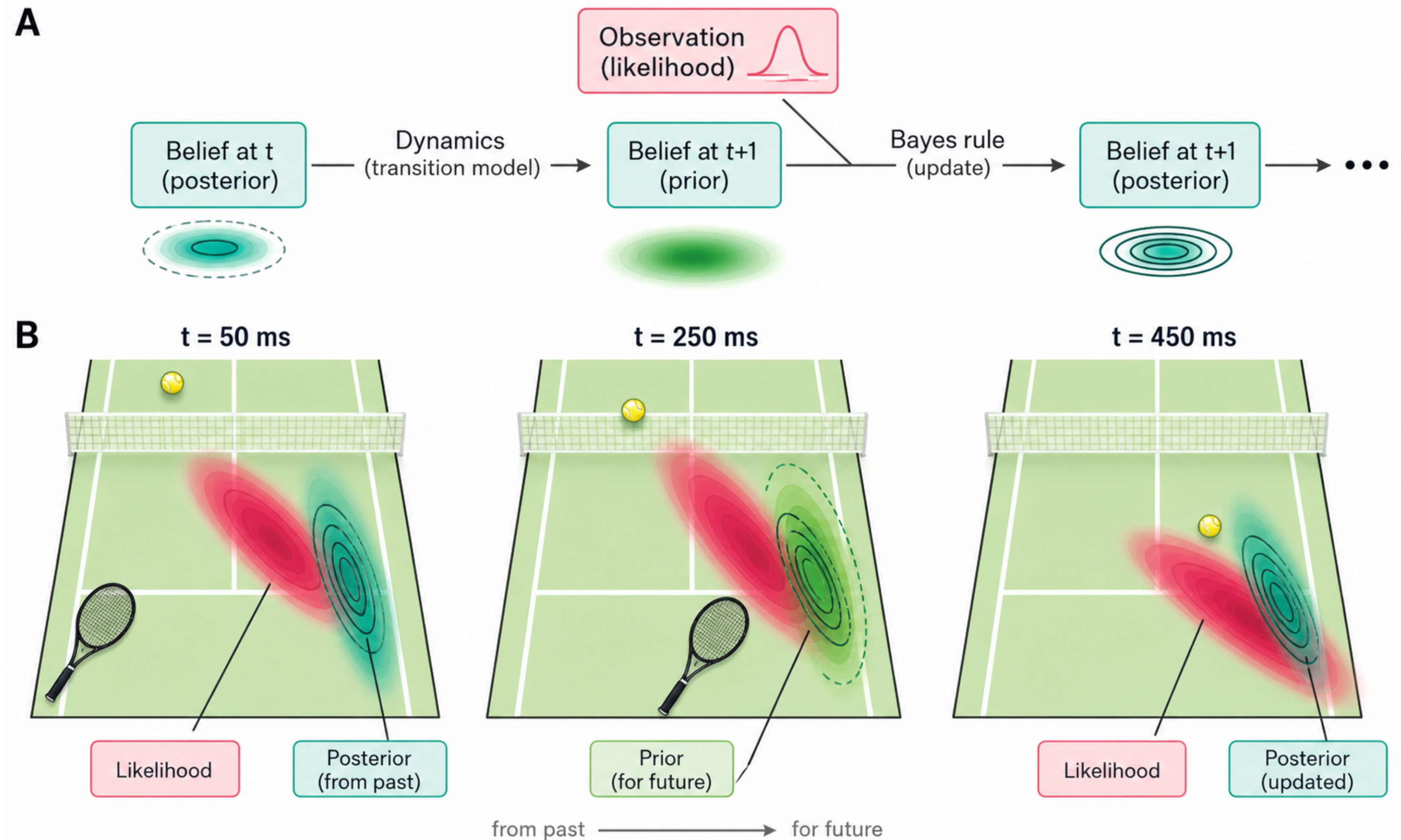


Very basics of Bayesian analysis

Jorge Lillo-Box



The principles of Bayesian inference

- ✓ **Probability statements** are not limited to data, but can be made **for model parameters and models themselves.**

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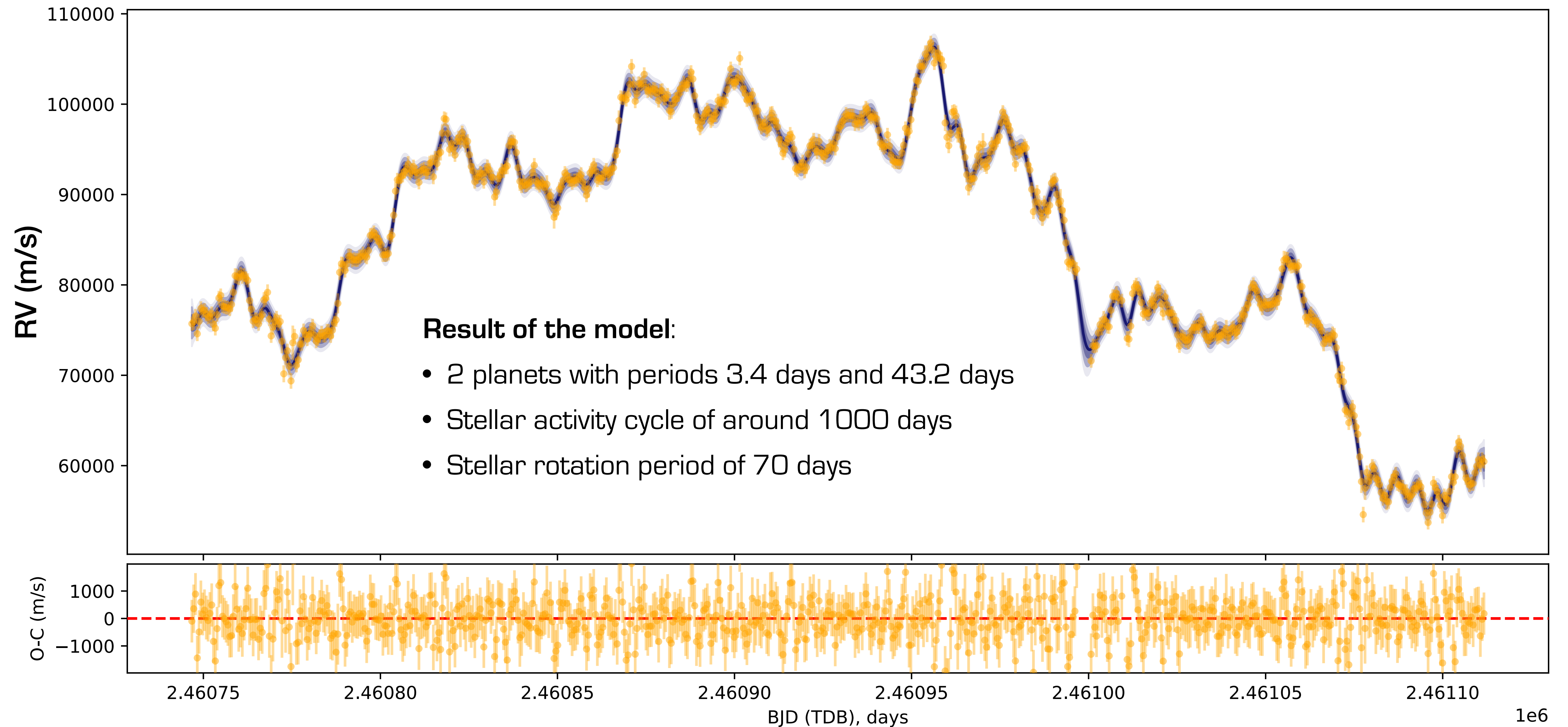
The principles of Bayesian inference

- ✓ **Probability statements** are not limited to data, but can be made **for** model parameters and models themselves.
- ✓ Inferences are made by producing **probability density functions** (PDFs)
- ✓ Model **parameters** are treated as **random variables**

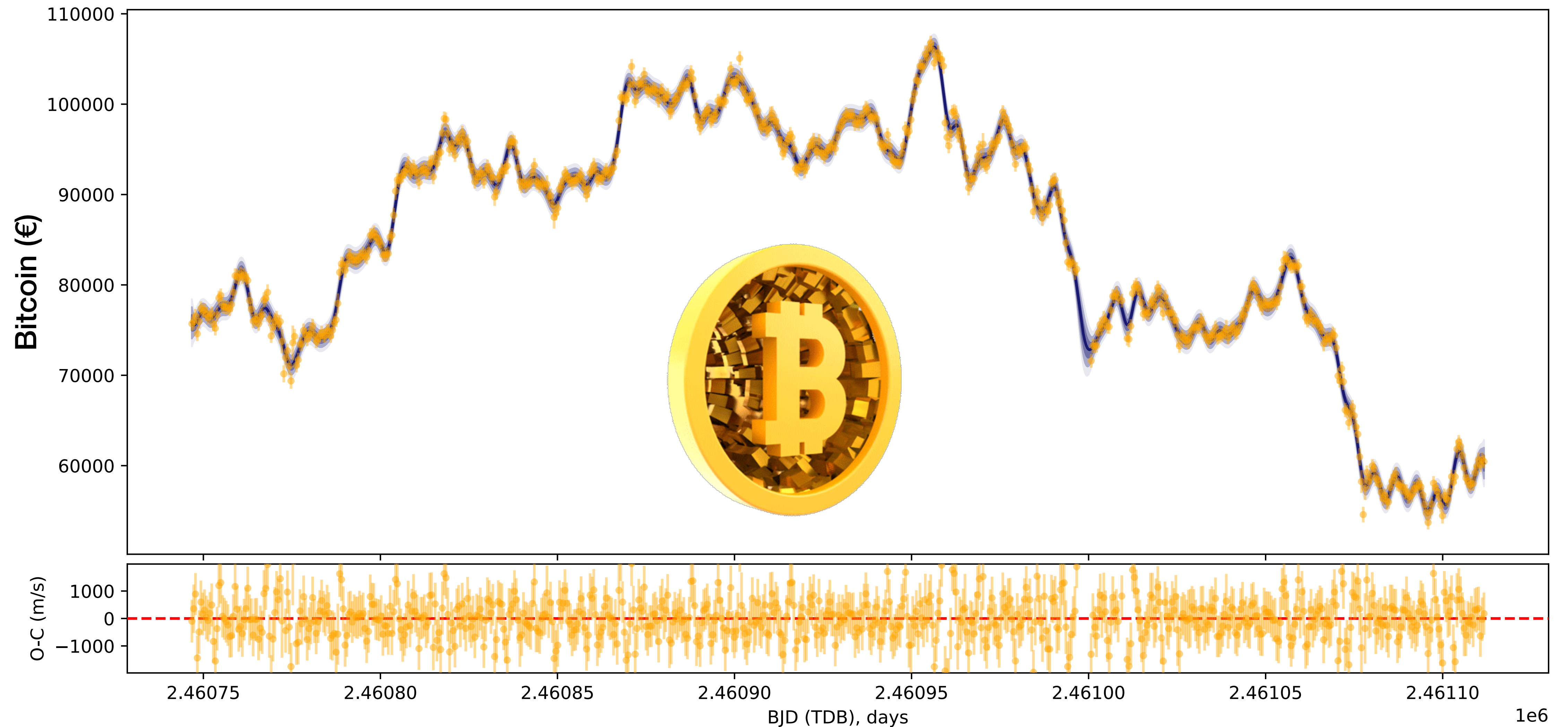
The principles of Bayesian inference

- ✓ **Probability statements** are not limited to data, but can be made for model parameters and models themselves.
- ✓ Inferences are made by producing **probability density functions** (PDFs)
- ✓ Model **parameters** are treated as **random variables**
- ✓ Remember, Bayesian method yields optimum results **assuming that all of the supplied info is correct (!!!)**

The principles of Bayesian inference



The principles of Bayesian inference



The principles of Bayesian inference

Classical statistics

Likelihood is used to find model parameters that yield the **highest** data likelihood



It **cannot be interpreted as a probability density function** for model parameters

Bayesian Methods

Extends the concept of the data likelihood function by **adding extra information** to the analysis (so-called prior)



One can **assign probability density function to model parameters**

Bayes' rule

$$\begin{array}{c} \text{Probability of} \\ \text{y given x} \end{array} p(y|x) = \frac{\begin{array}{c} \text{Probability of} \\ \text{x given y} \end{array} p(x|y) \begin{array}{c} \text{Probability} \\ \text{of y} \end{array} p(y)}{\begin{array}{c} \text{Probability of x} \end{array} p(x)}$$

Bayes' rule is not controversial, it is just mathematics

Bayes' theorem

The Frequentist vs. Bayesian **controversy*** sets in **when we apply Bayes' rule to the likelihood function** to obtain Bayes' theorem:

$$\begin{array}{c} \text{Posterior probability} \\ p(M|D) = \frac{\overset{\text{Likelihood}}{p(D|M)} \overset{\text{Prior}}{p(M)}}{\underset{\text{Evidence}}{p(D)}} \end{array}$$

D = data
M = model

*The reliance on subjective "prior beliefs" is the most prominent critique. Different researchers might choose different priors, leading to different conclusions from the same data.

Bayes' theorem

The Frequentist vs. Bayesian **controversy** sets in **when we apply Bayes' rule to the likelihood function** to obtain Bayes' theorem:

$$\text{Posterior probability } p(M|D) = \frac{\text{Likelihood } p(D|M) \text{ Prior } p(M)}{\text{Evidence } p(D)}$$

D = data
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An "**improved belief**" is proportional to the product of an "**initial belief**" and the **probability that the "initial belief" generated the observed data**.

Bayes' theorem: experiment

Pick bowl #1 or bowl #2...

close your eyes...

pick a **random cookie** from that bowl...

Bayes' theorem: experiment

Did you pick a cream or a chocolate cookie?

Evidence: you picked a cream cookie

Hypothesis: you picked the cookie from bowl #1

Probability of picking a
cream cookie from bowl #1

Prior probability of
picking bowl #1

$$P(H | E) = \frac{P(E | H) \times P(H)}{P(E)}$$

Probability that you picked
the cookie from bowl #1
given that it is plain

Prior probability of picking a
cream cookie from any bowl

Bayes' theorem: experiment

Did you pick a cream or a chocolate cookie?

Evidence: you picked a cream cookie

Hypothesis: you picked the cookie from bowl #1

$$P(H|E) = \frac{\text{Probability of picking a cream cookie from bowl \#1} \times \text{Prior probability of picking bowl \#1}}{\text{Prior probability of picking a cream cookie from any bowl}} = 75\%$$

Probability that you picked the cookie from bowl #1 given that it is plain

Bayes' theorem: experiment

Did you pick a cream or a chocolate cookie?

Evidence: you picked a chocolate cookie

Hypothesis: you picked the cookie from bowl #1

Probability of picking a **chocolate**
cookie from bowl #1

Prior probability of
picking bowl #1

$$P(H | E) = \frac{P(E | H) \times P(H)}{P(E)}$$

Probability that you picked
the cookie from **bowl #1**
given that it is **chocolate**

Prior probability of picking a
chocolate cookie from any bowl

Bayes' theorem: experiment

Did you pick a cream or a chocolate cookie?

Evidence: you picked a chocolate cookie

Hypothesis: you picked the cookie from bowl #1

Probability of picking a **chocolate**
cookie from bowl #1

Prior probability of
picking bowl #1

$$P(H|E) = \frac{5/20 \times 1/2}{20/40} = 25\%$$

Probability that you picked
the cookie from **bowl #1**
given that it is **chocolate**

Prior probability of picking a
chocolate cookie from any bowl

Bayes' theorem applied

$$\overset{\text{Posterior probability}}{p(M, \theta | D, I)} = \frac{\overset{\text{Likelihood}}{p(D | M, \theta, I)} \times \overset{\text{Prior}}{p(M, \theta | I)}}{\underset{\text{Evidence}}{p(D | I)}}$$

D = data

M = model

$\theta = \theta_1, \dots, \theta_k$ parameters of the model

I = any other information

$$p(D | I) = \int p(D | \theta) \times P(\theta) d\theta$$

Parameter space → normalization term, irrelevant in parameter inference, but critical in Model comparison

Bayes' theorem applied

$$\overset{\text{Posterior probability}}{p(M, \theta | D, I)} \propto \overset{\text{Likelihood}}{p(D | M, \theta, I)} \times \overset{\text{Prior}}{p(M, \theta | I)}$$

D = data

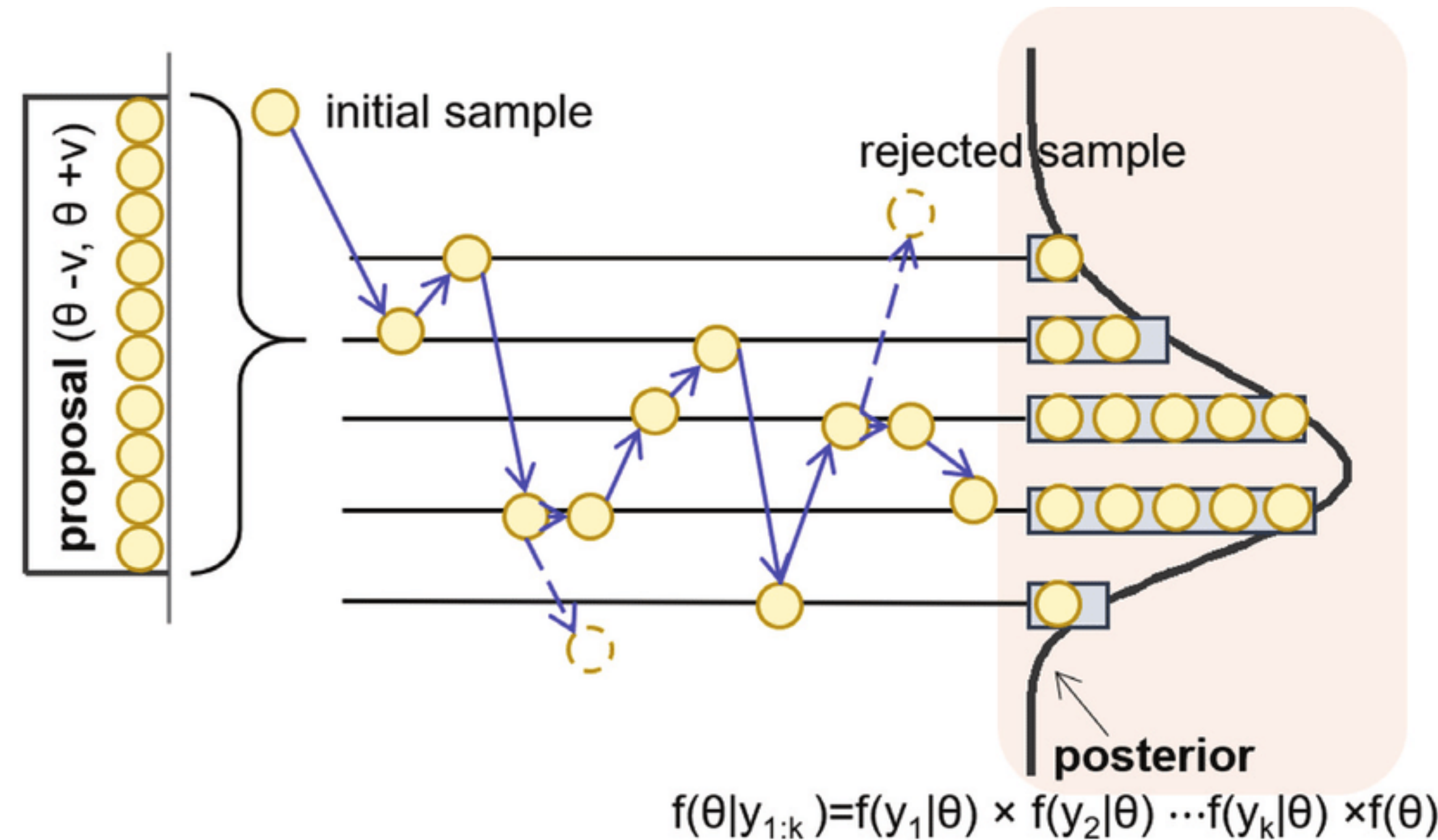
M = model

$\theta = \theta_1, \dots, \theta_k$ parameters of the model

I = any other information

These are the **three functions** we need to define to perform **Bayesian inference on our data**

Sampling the posterior distribution with MCMC



Posterior probability

Likelihood

Prior

$$p(M, \theta | D, I) \propto p(D | M, \theta, I) \times p(M, \theta | I)$$